Robust to What? Interpreting Shifted Success in Executable Vision-Language-Action Policies

Zihuan Wang, Chi Bene Chen, and Gao Huang*

***Abstract*— Terminal success under distribution shift is commonly treated as a checkpoint-level robustness score for vision-language-action (VLA) policies. This interpretation mistakes a marginal rate for a relation between clean and event-conditioned outcomes. The event marginal combines successes retained from clean execution with event-only successes, while hiding clean successes that are lost under the event. Identical event success rates can therefore correspond to different retention and loss structures. Moreover, the same frozen checkpoint can produce different terminal outcomes when the observation-to-action schedule changes, even when the evaluation cases remain fixed. A scalar reported without its measurement design therefore does not specify which reliability relation has been measured. We formalize terminal-success evaluation by $\mathcal{M} = (e, r, q)$, comprising the event, reference construction, and execution protocol. The framework distinguishes marginal comparison, state-index-matched outcome overlap, and trajectory-matched clean/event transitions. We show that clean and event marginals do not identify the paired transition vector, and that reference construction determines which of these outcome relations an evaluation can support. Across four public VLA systems and five simulated manipulation tasks, similar marginal scores conceal different observed overlap or paired transition structures. For the two 50-step systems, changing q also changes terminal outcomes for fixed checkpoints and evaluation cases. These results establish that terminal success is not a context-free property of a checkpoint. VLA evaluations should therefore specify e, r, and q, align the reference construction with the intended reliability claim, and report transition counts and effective denominators whenever the design supports them.**

Keywords—Vision-language-action policies, reliability evaluation, distribution shift, matched intervention, execution protocol.

I. Introduction

What reliability claim does a terminal success rate support when a vision-language-action (VLA) policy is evaluated under distribution shift? A shifted-condition success rate is often treated as a checkpoint-level robustness score. Yet the same scalar can reflect limited nominal competence, retained success, clean-only outcomes, or event-only success. The same checkpoint can also produce different endpoint rates under different observation-to-action schedules. These cases require different evidence. A ranking by the scalar therefore does not identify a ranking by outcome retention, event-associated loss, or response under a specified execution protocol.

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The ambiguity follows from the statistic itself. For matched reference and event outcomes $Y_0, Y_e \in \{0, 1\}$,

$$\Pr(Y_e = 1) = \Pr(Y_0 = 1, Y_e = 1) + \Pr(Y_0 = 0, Y_e = 1). \tag{1}$$

The event marginal combines retained and event-only successes while omitting clean-only outcomes. It is therefore a many-to-one projection of the four paired transitions $(p_{11}, p_{10}, p_{01}, p_{00})$, where $p_{ab} = \Pr(Y_0 = a, Y_e = b)$. Similar marginal scores can consequently represent different outcome-transition structures and support different outcome-level claims.

Existing benchmarks address parts of this ambiguity. Perturbation suites such as LIBERO-Plus broaden the visual, spatial, linguistic, and sensor perturbations evaluated [1]. Progress-aware benchmarks and failure analyses enrich the description of what occurs within an episode beyond binary task completion [2], [3], [4]. These advances improve perturbation coverage and outcome reporting. However, neither broader coverage nor richer reporting by itself specifies how cases are compared or when a changed observation can affect control. We therefore ask what terminal success identifies when the event, reference, and opportunity to respond have not all been specified.

Answering this question requires specifying what the measurement compares and how the policy is executed. We therefore index each terminal-success measurement by $\mathcal{M} = (e, r, q)$:

- **Event (e): what changed and what response the evaluation requires.** An appearance-only event preserves the task-relevant physical state, so the relevant requirement is outcome retention. An event that changes the controlled object state may instead require a new action.
- **Reference (r): which outcome relationships are observable.** Separate cohorts permit only a descriptive contrast between marginals. State-index matching reveals observed outcome overlap. Trajectory-matched branches additionally share the task, initial state, policy seed, event time, and execution protocol. Their post-event trajectories may diverge, but their terminal outcomes remain paired by construction.
- **Execution protocol (q): whether and when the updated observation can influence control.** The protocol specifies the observation query schedule and action-queue handling. We use *response opportunity* to denote when the updated observation reaches the next model query under q.

These components are not merely reporting details. To-

gether, they determine the statistical object identified by terminal success and, consequently, the strongest reliability claim supported by the evaluation.

We evaluate this claim using four public pretrained VLA systems across five simulated manipulation tasks. Figure 1 summarizes the three studies. The static study contains 1,600 rollouts, including 400 state-index-matched clean/dark outcome pairs. Although $\pi_{0.5}$ and OpenVLA-OFT both show a $+2$-percentage-point dark-minus-clean contrast, their clean-only/dark-only counts are $4/6$ and $11/13$ (Fig. 2). In 200 native trajectory-matched clean/intervention pairs, SmolVLA and OpenVLA-OFT obtain similar intervention success rates, $21/50$ and $23/50$, but clean-only/intervention-only counts of $17/0$ and $6/8$ (Fig. 3). The protocol control replays 100 of these system–key pairs under clear-and-re-query. It changes the clean/intervention transition vectors for both 50-step systems and raises intervention success by 16 and 30 percentage points, compared with 4 and 2 points in the corresponding clean branches. This establishes dependence on q without isolating stale-action latency as the causal mechanism.

Together, these measurements distinguish clean-condition performance, event-associated outcome transitions, and dependence on the execution protocol. They do not by themselves localize perception, replanning, or stale-action mechanisms; instead, they determine which follow-up audit is warranted. Runtime assurance and resilient planning for autonomous intelligent machines require this distinction because an endpoint failure alone cannot determine whether to inspect clean-condition task performance, the response to the event, or the observation-to-action interface.

Our contributions are:

1) We formulate terminal-success evaluation through the measurement specification $\mathcal{M} = (e, r, q)$ and its identified summary $\theta_{\mathcal{M}}$. Equation (1) and Table I formalize which outcome-level claim each evaluation design can support.
2) Using 1,600 static rollouts and 200 native trajectory-matched clean/intervention pairs, we show that identical or similar marginals can conceal different observed overlap or transition structures (Figs. 2 and 3). The analysis explicitly separates state-index matching from trajectory matching.
3) Using clear-and-re-query replays of 100 system–key pairs, we show that response opportunity is part of the evaluated execution protocol. Table IV reports changes in both the clean/intervention transition vectors and intervention success, establishing protocol dependence without claiming an isolated latency mechanism or a recovery method.

II. Related Work

Perturbation coverage. SIMPLER evaluates whether simulated policy performance and sensitivity to distribution shifts reflect paired real-world results [5]. LIBERO provides language-conditioned manipulation tasks, demonstrations, and reproducible initial states [6]. LIBERO-Plus expands this setting to layout, viewpoint, robot initialization, language, lighting, texture, and sensing perturbations, with progressive difficulty levels [1]. Its central question is where and by how much VLA performance degrades. Our study uses fewer events and does not compete on coverage. It asks a different question: whether a condition-level success rate identifies outcome retention or loss for specified cases. Broader perturbation coverage and better interpretation of each reported statistic are complementary requirements.

Outcome metrics and failure analysis. BEHAVIOR complements task completion with predicate-level progress and efficiency measures, including normalization to human demonstrations [2]. CALVIN evaluates long-horizon language-conditioned manipulation through completion along five-instruction chains rather than only isolated-task success [3]. REFLECT and its RoboFail dataset instead evaluate the localization, explanation, and correction of execution and planning failures [4]. These works enrich what can be measured or explained within an execution. We study a different omission: a shifted marginal collapses the four outcome transitions between matched clean and shifted branches. Our paired reference therefore addresses cross-condition transition identification rather than progress scoring, sequence-level completion, or failure explanation.

Recovery methods and executable interfaces. FailSafe, FLARE, and Affordance Field Intervention develop mechanisms that detect failure or select corrective behavior [7], [8], [9]. Deployment analysis also shows that normalization, observation construction, action decoding, and controller conventions are part of an executable VLA policy [10]. Receding-horizon visuomotor policies commonly predict action sequences [11]. Recent VLA-specific work also treats action chunks and control frequency as explicit choices in the action-tokenization interface [12]. VLA-Corrector makes the resulting responsiveness trade-off explicit and addresses it algorithmically by detecting drift, truncating stale actions, and triggering corrective replanning [13]. We do not propose queue clearing as a recovery method. We use a fixed-trigger protocol comparison to test whether the outcome summary and clean/intervention transition structure change when the remaining action queue is cleared and the frozen checkpoint is queried immediately after the updated observation.

Taken together, the closest work broadens perturbation coverage, enriches within-episode outcome reporting, or modifies recovery and action execution. We instead examine how event specification, reference construction, and execution protocol constrain the claim supported by terminal success. Our contribution is a measurement analysis, not a new robustness benchmark or recovery controller.

III. Reliability Claims and Evaluation Protocol

A. A measurement indexed by event, reference, and execution protocol

We define a terminal-outcome measurement specification conditional on a fixed evaluation background. Let b collect the frozen checkpoint, task pool, terminal-success criterion,

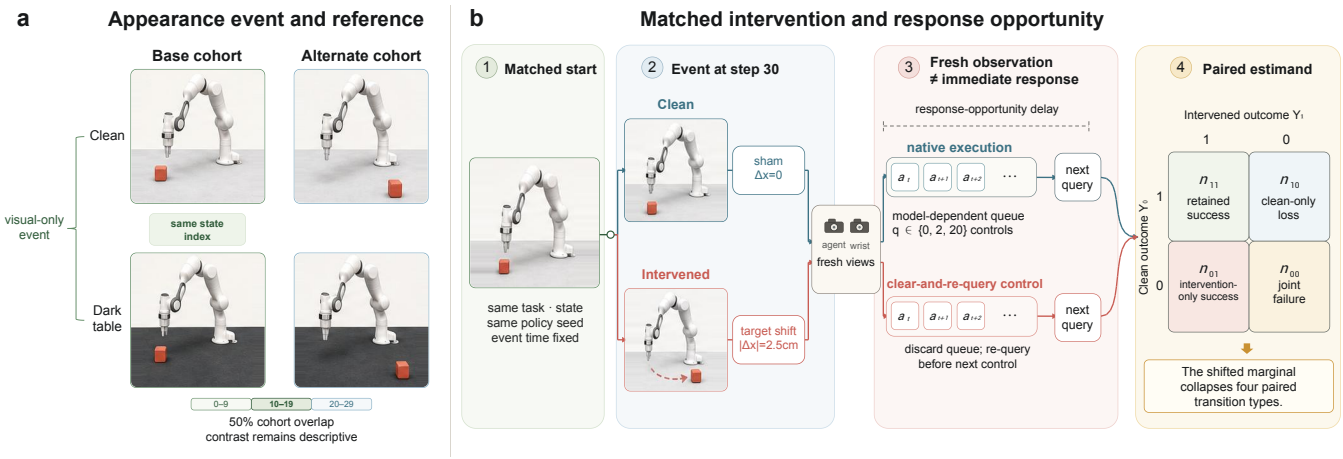


Fig. 1. Evaluation constructions for interpreting shifted success. (a) Static appearance comparison: clean and dark tables across base states 0–19 and alternate states 10–29. Their 10-state overlap is 50% per cohort, so the between-cohort contrast is descriptive; base-cohort clean/dark outcomes are matched by state index. (b) Fifty keys per checkpoint match the task, initial state, policy seed, and step-30 event time. The clean branch applies a zero-translation sham, whereas the intervened branch shifts the target by 2.5 cm; both freshly render the agent and wrist views. Native execution leaves 0, 2, or 20 controls before the next query; for the two 50-step systems, clear-and-re-query discards the remaining queue. The counts $(n_{11}, n_{10}, n_{01}, n_{00})$ distinguish retained success, clean-only loss, intervention-only success, and joint failure; the intervention marginal $(n_{11} + n_{01})/N$ does not recover this structure.

and horizon. The full specification is

$$\mathcal{M} := (b; e, r, q), \tag{2}$$

where e is the event, r is the reference construction, and q is the execution protocol. Because b is held fixed within each comparison, we use the abbreviated notation $\mathcal{M} = (e, r, q)$ throughout the paper.

For each stated finite set of evaluation cases, $\theta_{\mathcal{M}}$ denotes the outcome summary over all cases made available by r. It is therefore a finite-pool target. If only a subset of those cases is evaluated, $\widehat{\theta}_{\mathcal{M}}$ denotes the same summary computed on the evaluated subset. The studies below evaluate every case in their stated frozen pools, so their reported point summaries equal the corresponding finite-pool targets; the hatted success quantities in the Results denote their empirical proportions.

With only an event branch, $\theta_{\mathcal{M}}$ is the event marginal S_e. Separate cohorts support a descriptive contrast between marginals. State-index matching supports observed outcome-overlap counts. Trajectory matching supports a paired transition vector. We call branches *trajectory-matched* when they share the task, initial state, policy seed, event time, and execution protocol; their post-event trajectories are allowed to diverge.

For a trajectory-matched reference over a finite pool $\mathcal{P}_r$ with $N_r = |\mathcal{P}_r|$, let $Y^q_{0,j}$ and $Y^q_{e,j}$ denote the clean and event outcomes for key j. We define

$$\mathbf{T}_{\mathcal{M}} = (p_{11}, p_{10}, p_{01}, p_{00}), \tag{3}$$

where

$$p_{ab} = \frac{1}{N_r} \sum_{j \in \mathcal{P}_r} \mathbf{1}[Y^q_{0,j} = a, Y^q_{e,j} = b]. \tag{4}$$

The corresponding marginals are $S_0 = p_{11} + p_{10}$ and $S_e = p_{11} + p_{01}$. The clean-conditioned event success is

$$S_{e|0=1} = \frac{p_{11}}{p_{11} + p_{10}}. \tag{5}$$

For the following result, a paired quantity is identified by the reported marginals only if every transition vector compatible with those marginals gives the same value.

Proposition 1 (Marginals do not identify paired transitions). Fix b, e, and q. Suppose that only the aggregate clean and event marginals are available for the same N_r comparison cases. Let $S_0 = a$ and $S_e = s$. Every paired transition vector compatible with these marginals has

$$\begin{aligned} p_{11} &= t, & p_{10} &= a - t, \\ p_{01} &= s - t, & p_{00} &= 1 - a - s + t, \end{aligned} \tag{6}$$

where t ranges over the feasible multiples of $1/N_r$ satisfying

$$\max\{0, a + s - 1\} \le t \le \min\{a, s\}. \tag{7}$$

Whenever this interval contains more than one feasible value, the marginals do not uniquely determine either p_{10} or $S_{e|0=1} = t/a$ for $a > 0$.

Proof. Setting $p_{11} = t$, the two marginal equations determine the other three entries as shown above. Requiring all four entries to be nonnegative gives the stated bounds. Different feasible values of t change the clean/event outcome overlap, and therefore change p_{10} and $S_{e|0=1}$, while leaving both marginals unchanged. □

An event marginal alone contains still less information because S_0 is not available. Separate cohorts additionally do not preserve the same comparison cases. A per-case reference is therefore required to determine the four observed outcome cells, and trajectory matching is additionally required to interpret those cells as trajectory-level transitions. Thus r

TABLE I

TERMINAL-OUTCOME CLAIMS SUPPORTED BY THE EVALUATION DESIGNS USED IN THIS PAPER. STATE-INDEX MATCHING IDENTIFIES OBSERVED OVERLAP, TRAJECTORY MATCHING IDENTIFIES PAIRED TRANSITIONS, AND VARYING q IDENTIFIES PROTOCOL DEPENDENCE.

Evaluation design	**Supported terminal-outcome claim**	**Not identified from terminal outcomes**
Event branch only	S_e: success rate on the specified event-condition cases	Retention, event-associated loss, or event-only success
Clean/event cohorts summarized only by their marginals	Descriptive contrast between the sampled clean and event marginals	Per-case clean/event outcome relation
State-index-matched archived runs	Observed four-cell clean/event outcome overlap	Trajectory-level transitions under common stochastic replay
Trajectory-matched clean/event branches	Paired transition vector $\mathbf{T}_{\mathcal{M}}$ and clean-conditioned event success $S_{e\|0=1}$	Interpretation of intervention-only success as recovery
Same trajectory-matched cases, varied execution protocol q	Dependence of terminal outcomes and $\mathbf{T}_{\mathcal{M}}$ on q	Attribution of the change to a single component of q

is not merely a reporting detail: it determines which cross-condition outcome relationships are observable.

For an appearance-only event, $S_{e|0=1}$ describes outcome retention. For a physical intervention that changes the object state, it is clean-conditioned intervention success and should not be interpreted as an aggregate recovery rate.

When branches share only the official initial-state index and common stochastic replay is not certified, the same four-cell arithmetic describes observed outcome overlap on the archived runs. It does not identify trajectory-level transitions under common stochastic replay.

The event determines the response required by the reliability claim. The reference determines the comparison support and whether paired transitions are observable. The execution protocol determines when observations are queried and how queued controls are executed. We use response opportunity to denote when an updated observation reaches the next model query under q. Changing q changes the executable policy and may change $\theta_{\mathcal{M}}$ without identifying a latent failure mechanism. Table I serves as a reading guide: it links the evidence available in each evaluation design to the strongest conclusion that can be drawn from terminal outcomes.

In our experiments, the static study corresponds to the second and third rows, the physical-intervention study to the fourth row, and the clear-and-re-query comparison to the final row. The first row represents the event-condition marginal that would be reported if these additional comparisons were omitted.

B. Checkpoints and executable interfaces

We evaluate $\pi_{0.5}$, SmolVLA, OpenVLA, and OpenVLA-OFT [14], [15], [16], [17] on five spatially related bowl-to-plate tasks (task IDs 0, 1, 3, 7, and 9). All systems control the benchmark's relative end-effector controller at 20 Hz after ten no-op settling steps.

Table II records the frozen executable interfaces. Unless listed separately, each observation entry applies to both the static and intervention evaluations. In the execution column, each parenthesized value gives the number of controls remaining before the next model query after the step-30 event.

OpenVLA executes one decoded action per query, OpenVLA-OFT executes eight-step chunks, and $\pi_{0.5}$ and SmolVLA execute 50-step chunks. Because the event occurs at step 30 rather than at a common query boundary, the native post-event waits are 0, 2, or 20 controls. The native study therefore compares complete executable systems, not weights in isolation.

TABLE II

FROZEN EXECUTABLE INTERFACES FOR THE STATIC (S) AND PHYSICAL-INTERVENTION (I) EVALUATIONS.

Checkpoint	Observation interface	Native execution (wait)	Horizon (S/I)
$\pi_{0.5}$	2 RGB + state	50-step flow chunk (20)	280/220
SmolVLA	2 RGB + state	50-step flow chunk (20)	280/220
OpenVLA	1 RGB	Single token-decoded action (0)	220/220
OpenVLA-OFT	Static: 1 RGB Intervention: 2 RGB + state	8-step continuous chunk (2)	220/220

Two frozen protocol differences constrain cross-study interpretation. Static $\pi_{0.5}$/SmolVLA rollouts use a 280-step post-settle horizon, whereas their intervention rollouts use 220 steps. The OpenVLA-OFT static wrapper uses one camera without proprioception, while its intervention wrapper uses two cameras with proprioception. We consequently analyze static and physical-intervention experiments separately and do not treat their values as controlled within-interface effects.

C. Static event and partially overlapping cohort control

For each system m and task k, the static study evaluates a 2×2 table over initial-state cohort c and appearance a. The base cohort contains official states 0–19; the alternate cohort contains states 10–29. The cohorts overlap on states 10–19 and are not a matched pair. Each system–task–condition cell contains 20 episodes, yielding 1,600 rollouts.

The clean condition uses the default tabletop. The dark condition changes the tabletop and leg textures and materials while retaining the cohort. A structured XML audit finds identical bodies, geoms, collisions, cameras, lights, and physical attributes; all ten changed attributes belong to two textures and two table materials. "Dark" is therefore an

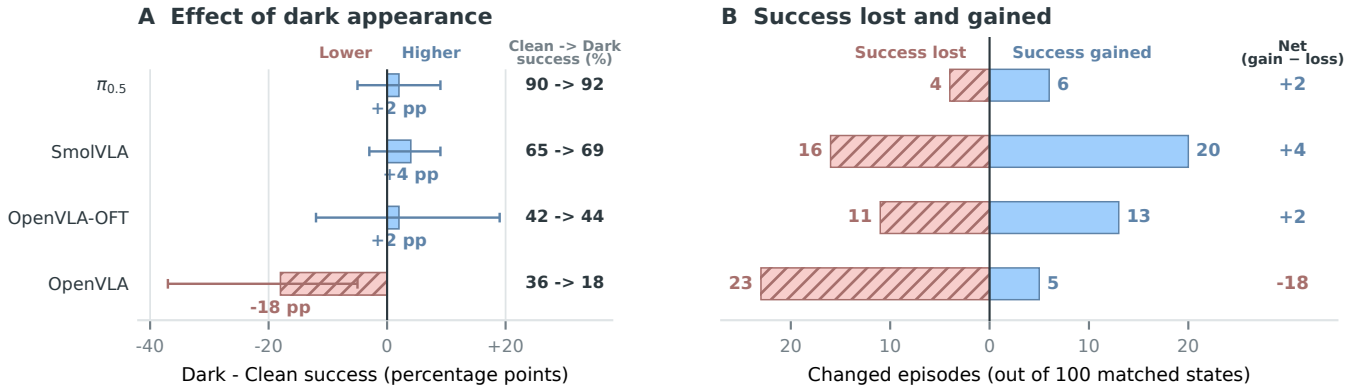


Fig. 2. Base-cohort dark-minus-clean success contrasts across five tasks are $+2$, $+4$, $+2$, and -18 percentage points for $\pi_{0.5}$, SmolVLA, OpenVLA-OFT, and OpenVLA, respectively. Only OpenVLA's descriptive task-bootstrap 95% interval lies wholly below zero. Panel A reports the dark-minus-clean contrasts and the corresponding clean-to-dark mean success rates. Bars are point estimates over the five observed tasks; horizontal whiskers are descriptive 95% intervals obtained by resampling those tasks, rather than population-level confidence intervals. Panel B decomposes the same 100 state-index-matched clean/dark outcome pairs per system into clean-only and dark-only counts; Net equals dark-only minus clean-only. SmolVLA's $+4$-point contrast, for example, combines 16 clean-only and 20 dark-only outcomes. Hatching marks clean-only outcomes, while solid bars mark dark-only outcomes. The displayed analysis uses 800 base-cohort rollouts (400 clean/dark pairs) from the 1,600-rollout static factorial study, with 20 episodes in each system–task–appearance cell. Because the static archive does not uniformly certify common action-noise replay, these counts describe state-index-matched outcome overlap rather than trajectory-level retention.

appearance-only event, not a global illumination or physical-state change.

Let $S_{mk}(c, a)$ denote terminal success. We report

$$\Delta_{\text{app}} = S(\text{base}, \text{dark}) - S(\text{base}, \text{clean}), \tag{8}$$

$$\Delta_{\text{cohort}} = S(\text{alt}, \text{clean}) - S(\text{base}, \text{clean}). \tag{9}$$

These are descriptive contrasts between marginal success rates. Δ_{app} measures terminal outcome contrast; it is not itself a retention estimand. Δ_{cohort} demonstrates the consequence of changing the reference cases; it is neither a paired event effect nor an online intervention. Task-level bootstrap intervals summarize heterogeneity across the five observed tasks and are not population-level confidence statements. The dark–alternate cell completes the factorial design but is not used to support an additional reliability claim.

For the base cohort, we also pair clean and dark outcomes by the same official initial-state index in the frozen raw logs. SmolVLA additionally records the same episode seed across each pair. The remaining static archives do not uniformly certify common policy-noise replay. We therefore call these *state-index-matched outcome pairs*, not trajectory-matched pairs. Their four-cell counts describe observed clean/dark outcome overlap on the archived runs; they do not identify trajectory-level retention. For the four prespecified system-level comparisons, two-sided exact McNemar tests compare clean-only and dark-only outcomes, with Holm correction across the four tests.

D. Trajectory-matched post-intervention outcomes

For each task, we freeze ten distinct official initial states selected by SmolVLA clean success in a separate screening run. The resulting 50-state pool, denoted $\mathcal{P}_{\text{Smol}}$, is reused unchanged for all checkpoints. SmolVLA is used only to construct this fixed common-state cohort: the pool is frozen before cross-checkpoint evaluation and is not reselected for any checkpoint. The design therefore supports comparison on common task–state–seed keys, not a model-neutral population ranking. Screening success does not force success in the formal paired replay; SmolVLA succeeds on 38/50 clean branches under the frozen intervention wrapper. Every task contributes ten distinct official initial states.

Each pair starts clean and intervened environments from the same official state and policy seed. After settling, both execute the same system until step 30. The intervened branch translates the target by $[-0.025, 0, 0]$ m in world coordinates, advances MuJoCo forward dynamics, and freshly renders the agent and wrist cameras. The clean branch performs the same forward and render calls with zero translation. The branches therefore differ in target displacement, not in whether fresh rendering occurs. All 200 checkpoint–key pairs pass the fresh-render integrity audit. Execution continues to terminal success or the 220-step horizon under the native query schedule.

We use the 2.5 cm, $-x$, step-30 displacement as a single operating point, chosen before the formal cross-checkpoint evaluation and held fixed across systems. This design identifies protocol-specific outcome transitions rather than a direction- or phase-averaged recovery quantity. The event is not normalized by object size, success tolerance, contact phase, or action-chunk boundary, and contact phase may differ across system–task pairs at step 30. The results therefore include cases in which the displacement makes the geometry easier as well as cases in which it disrupts progress.

Let n_{ab} count keys with clean outcome a and intervened outcome b. Primary evidence is the transition vector $\mathbf{n} = (n_{11}, n_{10}, n_{01}, n_{00})$. We additionally report

$$\widehat{S}_{\text{int}} = (n_{11} + n_{01})/N, \tag{10}$$

$$\widehat{S}_{\text{int}|c} = n_{11}/(n_{11} + n_{10}). \tag{11}$$

The conditional quantity is interpreted only with its clean-success denominator $n_{11} + n_{10}$. Wilson intervals are descrip-

TABLE III

MATCHED OUTCOMES UNDER CLEAN AND PHYSICAL-INTERVENTION CONDITIONS ON THE SHARED SMOLVLA-SCREENED POOL ($N = 50$ KEYS PER SYSTEM). BRACKETS SHOW DESCRIPTIVE WILSON 95% INTERVALS, p_{H} GIVES HOLM-ADJUSTED EXACT MCNEMAR VALUES, AND PAIRED COUNTS DISTINGUISH BOTH-SUCCESS, CLEAN-ONLY, INTERVENTION-ONLY, AND NEITHER OUTCOMES.

System	$\widehat{S}_{\mathrm{int}}$ [95% CI]	$\widehat{S}_{\mathrm{int\|c}}$ [95% CI]	C/I	p_{H}	B/C/I/N
$\pi_{0.5}$	40/50 [.67, .89]	37/45 [.69, .91]	8/3	.453	37/8/3/2
SmolVLA	21/50 [.29, .56]	21/38 [.40, .70]	17/0	$< .001$	21/17/0/12
OpenVLA-OFT	23/50 [.33, .60]	15/21 [.50, .86]	6/8	.791	15/6/8/21
OpenVLA	10/50 [.11, .33]	7/16 [.23, .67]	9/3	.438	7/9/3/31

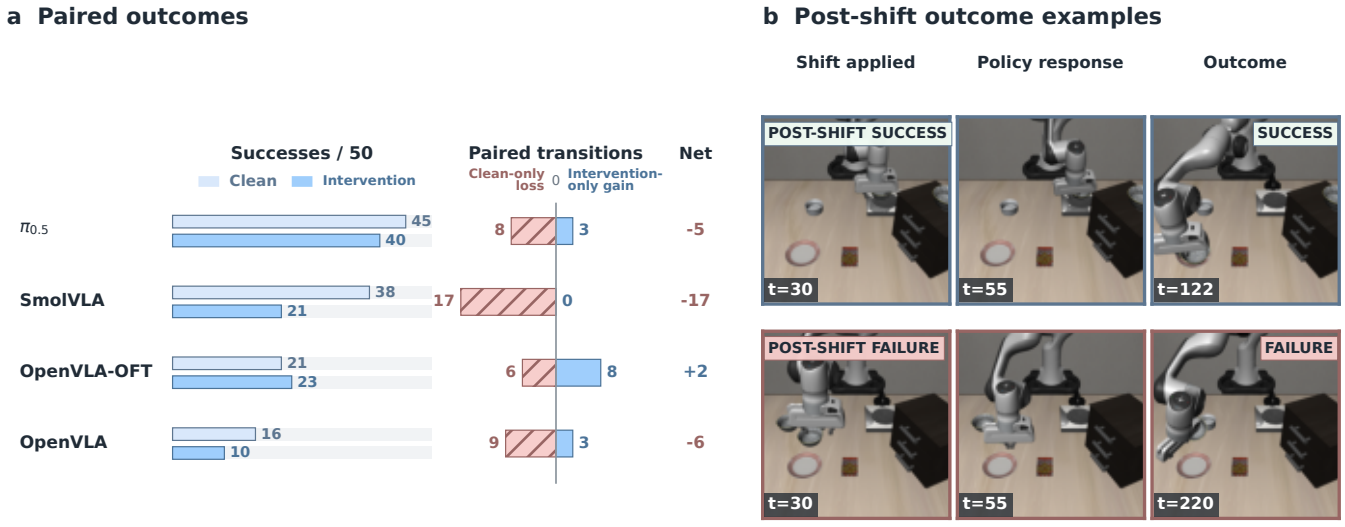


Fig. 3. Matched outcomes reveal structure hidden by marginal success rates. Left: clean and intervention success counts for four policies on the same SmolVLA-screened 50-key LIBERO-Spatial pool. The paired transitions report clean-only losses, intervention-only gains, and their net difference. The intervention translates the target object by 2.5 cm at step 30 while preserving the matched initial state and policy seed. Right: agent-view frames from one SmolVLA intervened branch ending in terminal success and one ending in terminal failure, shown immediately after the shift, during the subsequent policy response, and at the observed outcome. These deliberately selected trajectories illustrate outcome categories; they are not representative samples or an aggregate action-correction metric. Labels, borders, and hatch patterns preserve grayscale readability.

tive binomial intervals under an episode-level exchangeability approximation; the frozen-pool point estimates remain primary. Two-sided exact McNemar tests compare n_{10} and n_{01}; Holm correction covers the four prespecified system-level tests. For compactness, Table III uses $C/I = n_{10}/n_{01}$ for clean-only/intervention-only counts and B/C/I/N for $(n_{11}, n_{10}, n_{01}, n_{00})$; p_{H} denotes the Holm-adjusted McNemar value. Neither outcome summary identifies a directed corrective action without an additional behavioral criterion.

E. Fixed-trigger response-opportunity protocol control

To test whether the identified outcome summary depends on response opportunity, we replay the same 50 keys for each of the two 50-step policies under a second execution protocol. The native protocol q_{nat} retains the remaining action queue after the step-30 sham or physical intervention. The clear-and-re-query protocol q_{rq} discards the queue immediately after the fresh render and queries the frozen checkpoint before executing another control. The checkpoint, task, initial state, policy seed, event, reference, and horizon remain fixed.

This produces 100 additional clean/intervention pairs, or 200 additional rollouts, on the same 100 system–key units used by the native evaluation. OpenVLA and OpenVLA-OFT are not included because their native post-event waits are zero and two controls, respectively, rather than 20.

The protocol change modifies the executable policy. A fresh query may sample a different action chunk even in the sham-clean branch, so the comparison does not isolate stale-action latency as the sole mechanism. We report both the clean/intervention transition vector under each q and the native-versus- re-query discordance within the clean and intervened branches. In Table IV, $\mathbf{T}_q = (B, C, I, N)$ lists, in order, the both-success, clean-only, intervention-only, and neither counts under protocol q. Δ_i is the paired RQ-minus-native intervention-success difference, and Nat./RQ-only gives the corresponding discordant counts. The 95% CI for Δ_i uses 20,000 paired-bootstrap draws. Exact McNemar tests compare those discordant counts, with p_{H} denoting Holm-adjusted values across the two prespecified system comparisons.

IV. WHAT SHIFTED SUCCESS HIDES

We follow the identification map in Table I. The static study varies event and reference, the matched study exposes transition structure, and the protocol control varies response opportunity while holding the checkpoint and event fixed.

TABLE IV

EXECUTION-PROTOCOL CONTROL ON 50 MATCHED KEYS FOR EACH 50-STEP SYSTEM. NATIVE AND CLEAR-AND-RE-QUERY (RQ) EVALUATE THE ZERO-TRANSLATION CLEAN AND 2.5-CM STEP-30 INTERVENTION BRANCHES. BRACKETS GIVE PAIRED-BOOTSTRAP 95% CIS FOR THE RQ-MINUS-NATIVE INTERVENTION-SUCCESS DIFFERENCE. RQ CHANGES BOTH THE PAIRED TRANSITION STRUCTURE AND INTERVENTION SUCCESS FOR BOTH SYSTEMS.

System	Native clean	RQ clean	Native interv.	RQ interv.	$\mathbf{T}_{\mathrm{nat}}$	$\mathbf{T}_{\mathrm{rq}}$	Δ_i [95% CI]	Nat./RQ-only	p_{H}
$\pi_{0.5}$	45/50	47/50	40/50	48/50	37/8/3/2	45/2/3/0	+.16 [.04, .28]	2/10	.0386
SmolVLA	38/50	39/50	21/50	36/50	21/17/0/12	32/7/4/7	+.30 [.12, .48]	5/20	.0082

A. Static marginals depend on event and reference

Figure 2 reports base-cohort clean success and the corresponding dark-table contrasts. Mean clean success is 90% for $\pi_{0.5}$, 65% for SmolVLA, 42% for OpenVLA-OFT, and 36% for OpenVLA. These clean-condition success rates provide the baseline for interpreting each appearance contrast.

The dark-minus-clean contrasts are $+2$, $+4$, $+2$, and -18 percentage points in the same system order. State-index-matched clean-success overlap is 86/90, 49/65, 31/42, and 13/36. The descriptive task-bootstrap intervals cross zero for the first three systems, while OpenVLA's interval is $[-37, -5]$ percentage points. On the frozen state-index schedule, only OpenVLA shows a Holm-adjusted excess of clean-only over dark-only outcomes ($p_{\mathrm{Holm}} = 0.00365$).

SmolVLA provides the central counterexample. Its $+4$-point dark-minus-clean contrast combines 16 clean-only and 20 dark-only outcomes, so 49 of its 65 observed clean successes also succeed in the dark branch. A positive contrast can therefore coexist with 36 discordant outcomes: 16 clean-only and 20 dark-only.

Task-level appearance responses are heterogeneous. OpenVLA's contrast ranges from -55 percentage points on task 0 to -10 points on tasks 3 and 9; OpenVLA-OFT ranges from $+35$ points on task 0 to -20 points on task 9. The task-level cohort contrasts also change sign. Because half of the cohort cases differ, these values describe sensitivity to the sampled cases rather than a paired event effect. The static results therefore do not define a checkpoint ranking independent of the event and reference construction.

B. Paired transitions change the outcome-level interpretation

Table III and Fig. 3 report the paired outcome experiment. $\pi_{0.5}$ has intervened success 40/50, but the transition vector shows 37 both-success, eight clean-only, three intervention-only, and two neither outcomes. Its clean-conditioned intervention success is therefore 37/45. SmolVLA has intervened success 21/50; all 21 occur among its 38 clean successes, while 17 matched clean successes are lost. The marginal does not reveal this absence of intervention-only outcomes or the 21/38 effective clean-success denominator.

OpenVLA-OFT demonstrates the opposite ambiguity. Its clean and intervened marginals are 21/50 and 23/50, which could suggest little change. The matched cases instead contain 15 both-success, six clean-only, and eight intervention-only outcomes. Those eight outcomes are compatible with easier geometry, but terminal outcomes do not identify the action-level mechanism. Calling 23/50 a recovery rate would conflate retained and intervention-only success.

The SmolVLA–OpenVLA-OFT comparison is a concrete counterexample to ranking by the intervention marginal. Their rates differ by only two episodes, yet their transition vectors are $(21, 17, 0, 12)$ and $(15, 6, 8, 21)$, respectively. SmolVLA has 17 clean-only outcomes and no intervention-only outcome, whereas OpenVLA-OFT has six clean-only and eight intervention-only outcomes. Similar marginal rates can therefore correspond to different paired outcome structures. This contrast does not explain an architectural mechanism; it shows why the marginal cannot license one.

Task-level results show substantial heterogeneity behind the system-level summaries. SmolVLA's clean-conditioned intervention success ranges from 1/8 on task 3 to 7/7 on task 1, while OpenVLA-OFT ranges from 0/4 on task 7 to 8/9 on task 3. Each stratum contains only ten keys, so these values are heterogeneity diagnostics rather than task-level effect estimates.

OpenVLA supplies 16 clean-successful cases, seven of which also succeed under intervention. Its 7/16 conditional estimate has a wide descriptive interval [.23, .67], while the matched transitions contain nine clean-only losses and three intervention-only gains. The number of clean-success cases also varies across tasks: task 1 has no clean success, whereas task 3 contributes eight clean-success cases. Effective clean-success denominators therefore range from 16 to 45 and must accompany any conditional percentage.

At this sample size, only SmolVLA shows evidence of an imbalance between clean-only and intervention-only outcomes. Raw McNemar values are 0.227, 1.53×10^{-5}, 0.791, and 0.146 in table order. After Holm correction, only SmolVLA retains evidence of an imbalance favoring clean-only outcomes ($p_{\mathrm{Holm}} = 6.10 \times 10^{-5}$). For the other systems, the paired imbalances are inconclusive; this is not evidence of equivalence or improvement. The intervals rely on the stated episode-level approximation. They do not include uncertainty from task sampling, SmolVLA-based key selection, or alternative policy seeds.

The storyboard adds qualitative context without replacing aggregate evidence. In the selected successful rollout, the displaced target enters the updated observation and the end effector later changes direction before terminal success. The selected failure does not reach terminal success before the horizon. These trajectories illustrate two individual outcomes;

they do not estimate aggregate recovery ability or identify an action-level mechanism. Neither example changes the outcome-level interpretation of the 200 checkpoint–key pairs.

C. Outcome transitions depend on the execution protocol

This control compares $\mathcal{M}_{\mathrm{nat}} = (e, r, q_{\mathrm{nat}})$ and $\mathcal{M}_{\mathrm{rq}} = (e, r, q_{\mathrm{rq}})$ while holding the checkpoint, task, reference keys, policy seeds, event, and horizon fixed. Table IV reports the identified summaries under both protocols.

For $\pi_{0.5}$, the clean/intervention transition vector changes from $(37, 8, 3, 2)$ under native execution to $(45, 2, 3, 0)$ under re-query. Intervened success increases from 40/50 to 48/50, a paired difference of $+0.16$ with a 95% paired-bootstrap interval of $[0.04, 0.28]$. Ten keys succeed only under re-query and two only under native execution ($p_{\mathrm{Holm}} = 0.0386$).

For SmolVLA, the transition vector changes from $(21, 17, 0, 12)$ to $(32, 7, 4, 7)$. Intervened success increases from 21/50 to 36/50, a difference of $+0.30$ $[0.12, 0.48]$, with 20 re-query-only and five native-only outcomes ($p_{\mathrm{Holm}} = 0.0082$). Clean success changes by only $+0.04$ for $\pi_{0.5}$ and $+0.02$ for SmolVLA. The protocol change therefore affects the clean/intervention transition structure, not only the intervention marginal.

The difference is heterogeneous across tasks. For task IDs 0/1/3/7/9, the $\pi_{0.5}$ intervened differences are $+0.10$, 0.00, $+0.40$, -0.20, and $+0.50$; SmolVLA gives $+0.50$, 0.00, $+0.70$, $+0.10$, and $+0.20$. Each task contains ten pairs, so these values are descriptive heterogeneity checks rather than task-level efficacy estimates. The -0.20 estimate for $\pi_{0.5}$ on task 7 shows that the observed difference is not positive in every evaluated task.

Nearly unchanged clean marginals also conceal cross-protocol discordance: $\pi_{0.5}$ has one native-only and three re-query-only clean outcomes, while SmolVLA has six and seven. A fresh query can therefore change a trajectory even without object translation. The control identifies dependence on the execution protocol and its response opportunity; it is not a causal estimate of stale-action latency, weight-level adaptation, or recovery-method benefit.

V. Implications and Scope

A terminal-success report should specify the measurement design $\mathcal{M} = (e, r, q)$ and the identified summary $\theta_{\mathcal{M}}$. Trajectory-matched evaluations should report the four outcome transitions and the effective clean-success denominator; state-index-matched evaluations should report observed outcome overlap rather than trajectory-level retention. Together, these records separate clean-condition performance, event-associated transitions, and dependence on q, preventing results under different reference constructions or execution protocols from being conflated. The identification result concerns what each design can support; the observed frequencies remain specific to the evaluated systems, tasks, events, and protocols.

VI. Conclusion

Terminal success under distribution shift alone does not identify VLA reliability. We formalize terminal-success evaluation as $\mathcal{M} = (e, r, q)$ and show that identical clean and event marginals can correspond to different paired transition vectors. Across four VLA systems, similar marginals conceal different observed overlap or transition structures. Varying q changes outcomes for the same checkpoint and evaluation cases. These results establish that marginal success is ambiguous, not how often this ambiguity occurs across systems, tasks, or events.

Evaluation design should follow the intended reliability claim. A marginal supports claims about performance on stated event-condition cases; observed outcome overlap requires a per-case reference; trajectory-level retention or loss requires trajectory matching; and protocol dependence requires a controlled comparison of q. Paired evaluations should report four transition counts and the clean-success denominator. These measurements direct the next audit toward nominal task performance, the response to the event, or the observation-to-action interface; localizing a failure mechanism requires behavioral evidence.

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